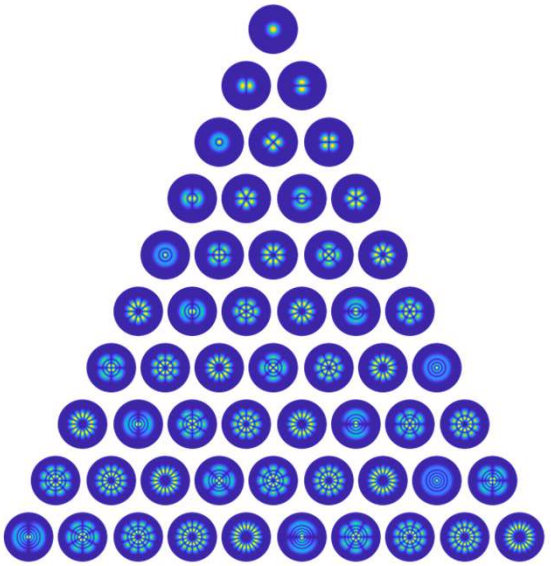
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2024/25

Lecture 2



Prof. Dr.-Ing. Georg Rademacher

Organizational matter :



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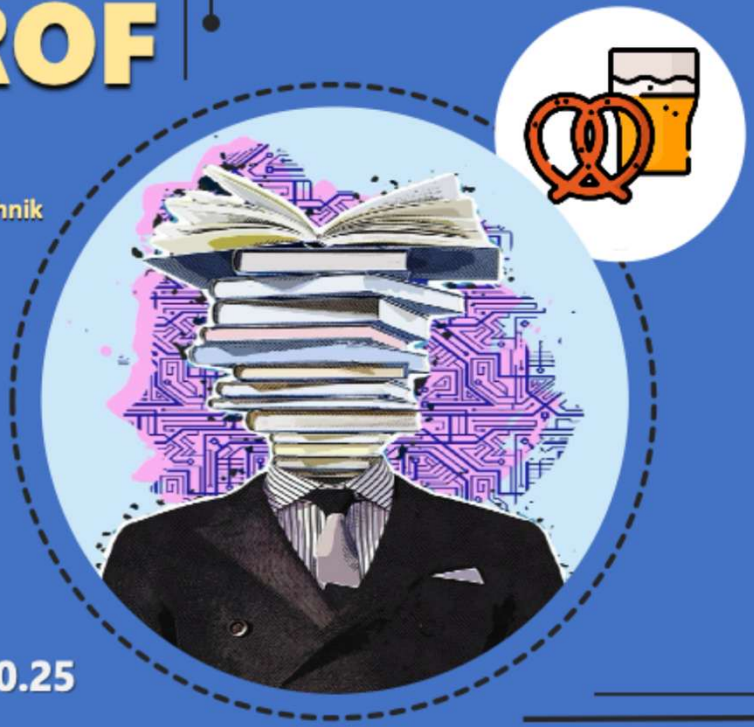
- This module contains a lecture and an exercise
- Typically, the lecture is held Thursdays at 2 pm in room (V 47-06)
- The exercise is on Tuesdays at 2 pm in this room (V 47-06) and will be held by Julian Schneck

- There won’t be a course reader (skript) but we will upload the slides from this lectures
- We recommend Govind P. Agrawal, Fiber Optic Communications Systems

MEET YOUR PROF

BIER- & BREZEL-ABEND*

Für Erstis des Bachelor-Studiengangs Elektrotechnik und Informationstechnik



13. November

17:30 Uhr

UG Pfaffenwaldring 47

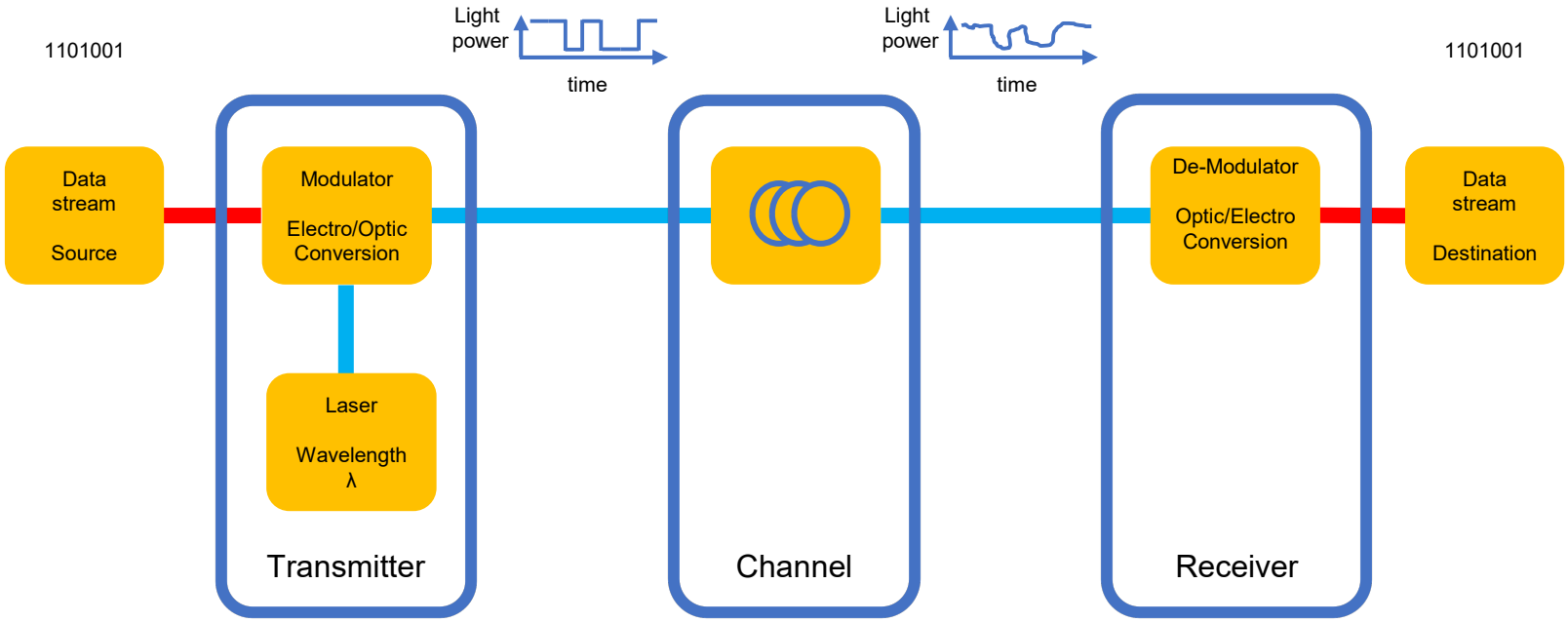
Anmeldung erfolgt über C@MPUS bis 16.10.25

Komm vorbei und lerne Deine Professoren und Professorinnen in lockerer Atmosphäre kennen!

* Bier, Brezeln sowie alkoholfreie Getränke sind gratis, solange der Vorrat reicht!

Principle of an optical fiber transmission link

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Plane waves

Plane waves



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Plane waves

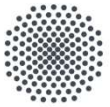


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Plane waves

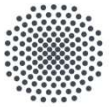
Plane waves

Plane waves



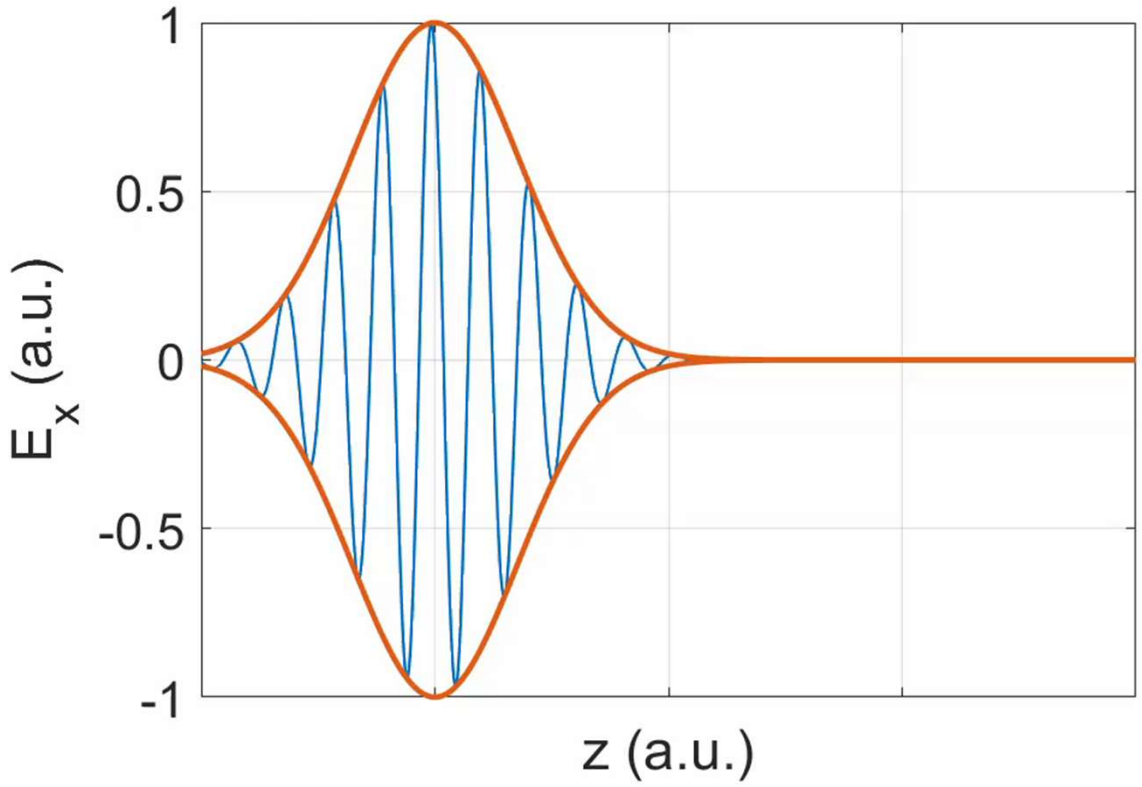
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Plane waves

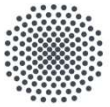


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Chromatic Dispersion



Chromatic Dispersion



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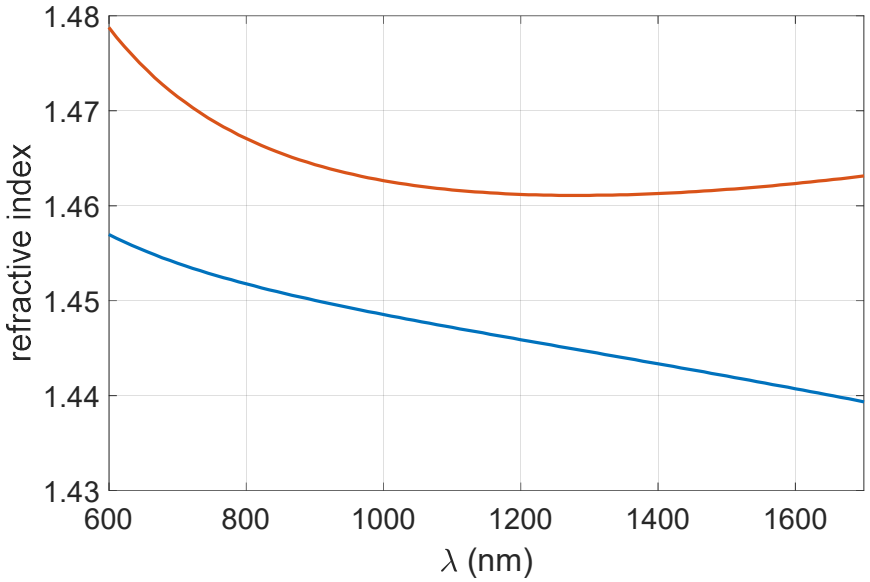
Chromatic Dispersion

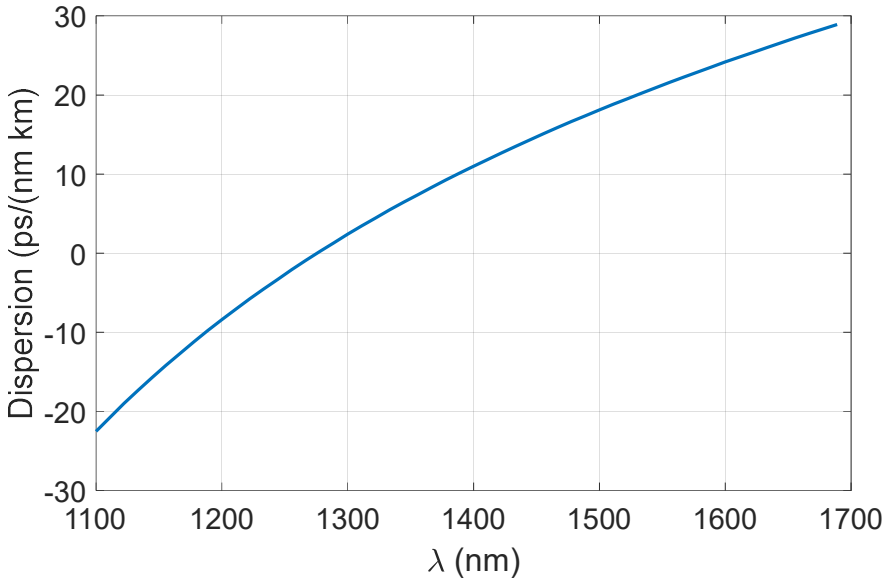


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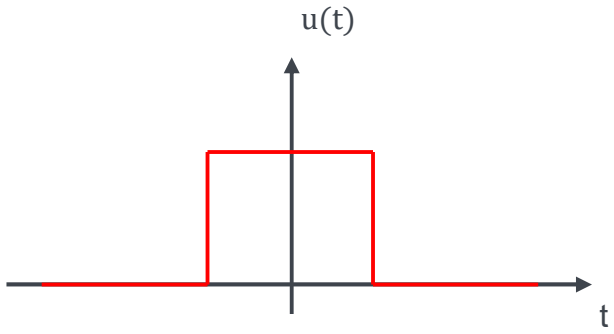
Chromatic Dispersion

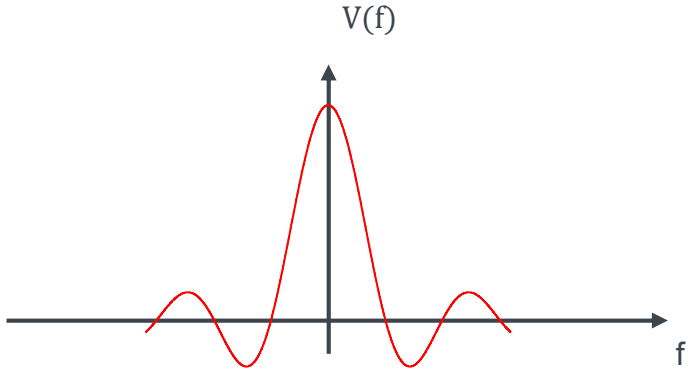
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Dispersion

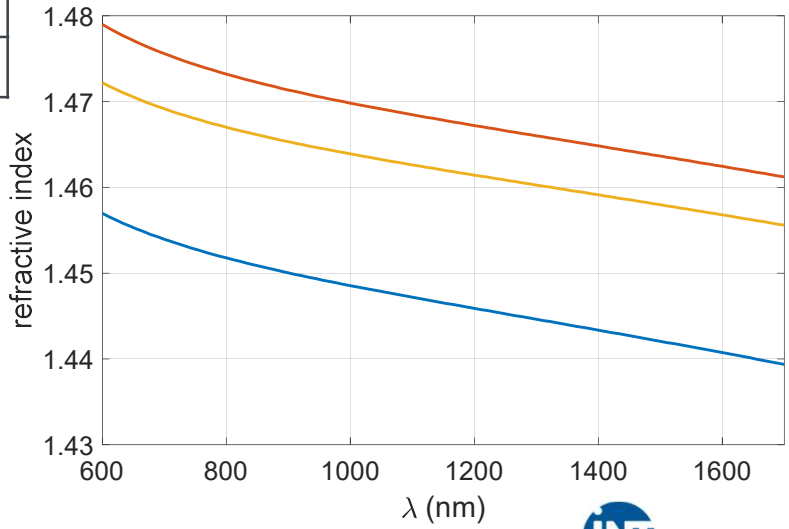




Refractive index with Sellmeier equations

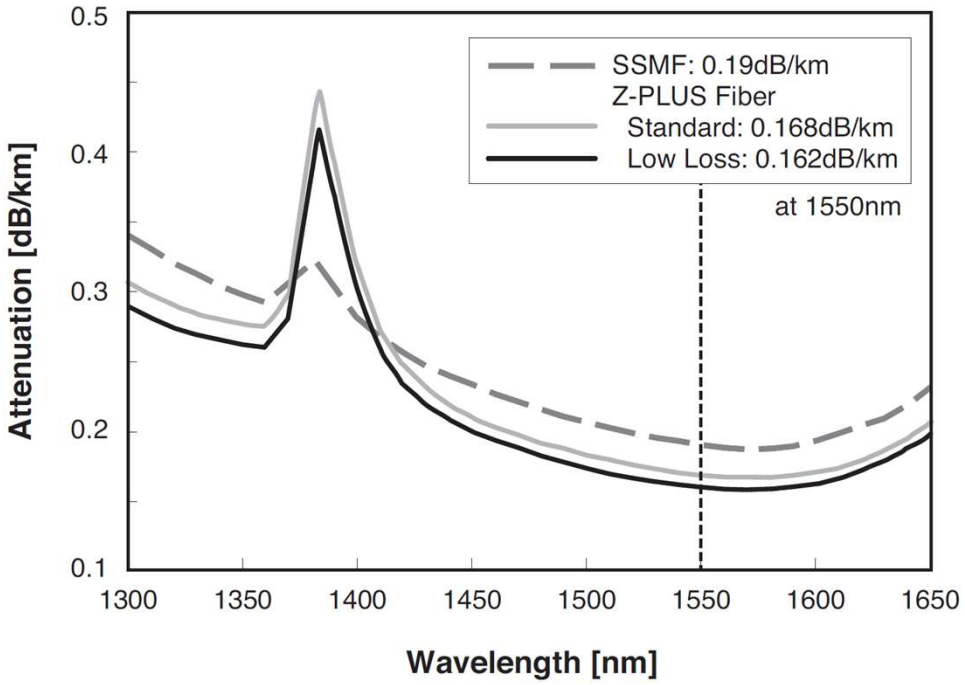
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Molar composition in %					f_e	f_d	f_L
SiO_2	B_2O_3	GeO_2	P_2O_5	Na_2O	in 10^{14} Hz		
100					32,12	35,31	0,31
86,7	13,3				31,27	34,20	0,33
86,5		13,5			30,21	34,99	0,29
90			10		32,05	36,61	0,29
58	19			23	30,53	39,34	0,31
18	58			24	30,98	39,27	0,36



Attenuation

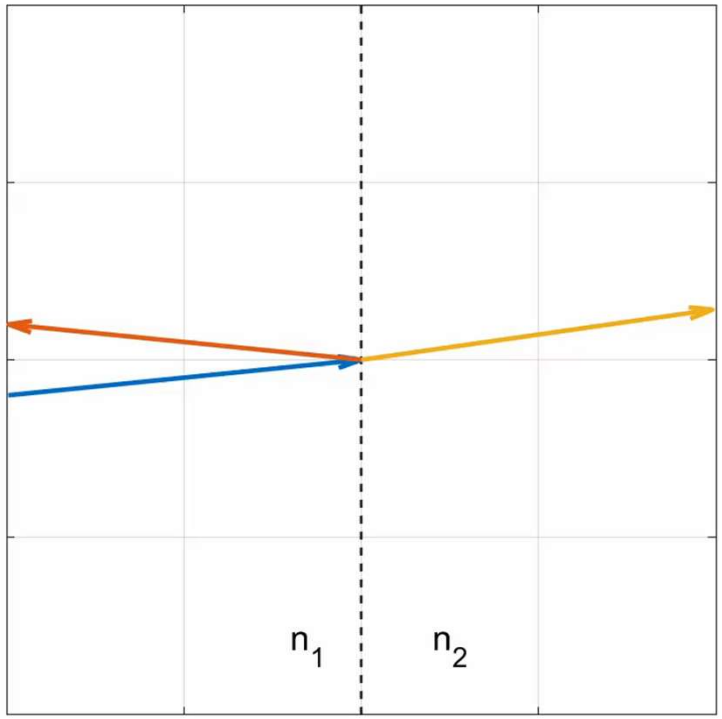
Attenuation



Yamamoto et al, Low-Loss and Low-Nonlinearity Pure-Silica-Core Fiber for Large Capacity Transmission, SEI Technical review, 2013

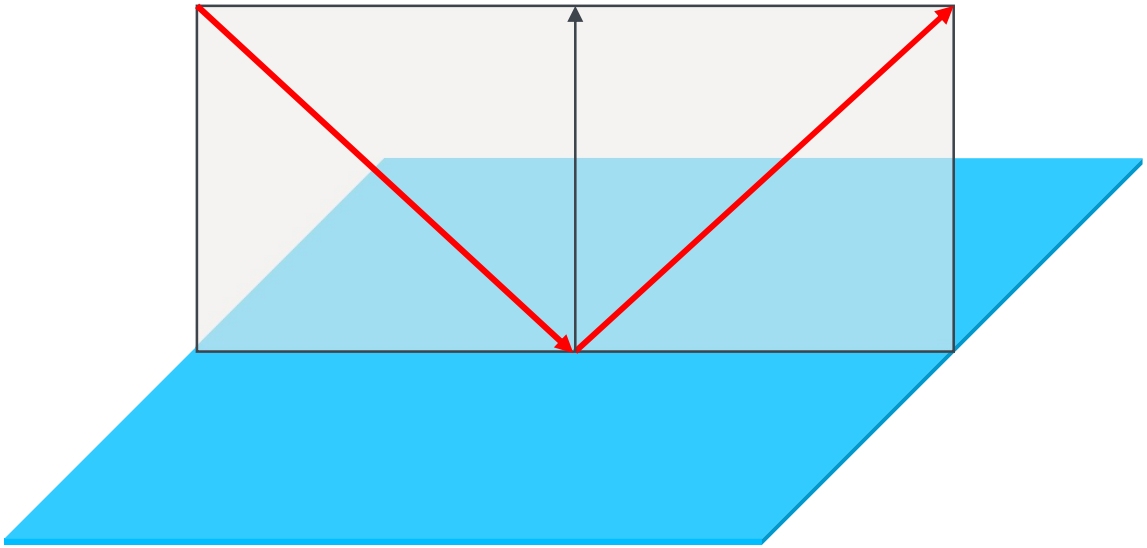
Snell's law



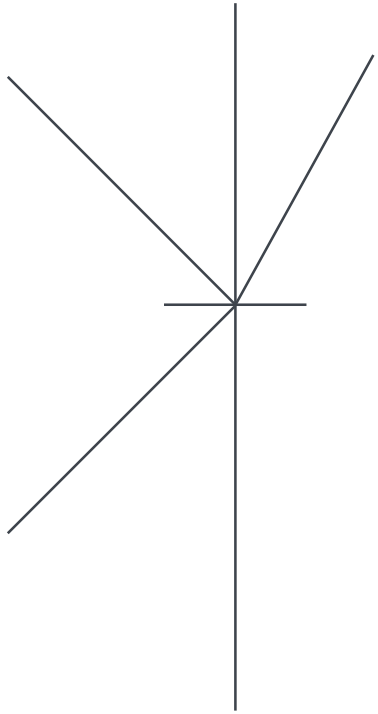


Plane of insidence

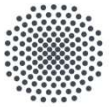
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Plane waves on dielectric interfaces



Plane waves on dielectric interfaces



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Reflectivity of plane waves on dielectric interfaces



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Plane waves on dielectric interfaces

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Institute of electrical and optical communications

Thank you!

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